



GODAVARI GLOBAL UNIVERSITY

B.Tech – Computer Science & Engineering

INDUSTRIAL INTERNSHIP REPORT

Data Science & Machine Learning – Agriculture Track

Prediction of Agriculture Crop Production in India & Crop and Weed Detection using YOLOv8

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Under the guidance of

Upskill Campus (USC)

in association with

UniConverge Technologies Pvt. Ltd. (UCT)

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Executive Summary

This report presents a comprehensive account of the 6-week Industrial Internship undertaken by Shaik Mahammad Shaheer under Upskill Campus in partnership with UniConverge Technologies Pvt. Ltd. (UCT). The internship focused on the Agriculture track within the Data Science and Machine Learning domain. Two real-world projects were undertaken: (1) Prediction of Agriculture Crop Production in India using regression models including Linear Regression, Random Forest, and XGBoost; and (2) Crop and Weed Detection using the YOLOv8 object detection framework. The internship provided hands-on exposure to the complete ML pipeline — from data collection and cleaning through EDA, model building, hyperparameter tuning, and evaluation — alongside practical experience in computer vision and deep learning workflows.

Table of Contents

1	Preface	3
2	Introduction	4
2.1	About UniConverge Technologies Pvt. Ltd.	4
2.2	About Upskill Campus (USC)	5
2.3	Objectives of this Internship	5
2.4	Reference	6
2.5	Glossary	6
3	Problem Statement	7
4	Existing and Proposed Solution	8
5	Proposed Design / Model	9
5.1	High Level Diagram	9
5.2	Low Level Diagram	9
5.3	Interfaces	9
6	Performance Test	10
6.1	Test Plan / Test Cases	10
6.2	Test Procedure	10
6.3	Performance Outcome	11
7	My Learnings	12
8	Future Work Scope	13

Preface

This report summarises six weeks of intensive work completed during my Data Science and Machine Learning internship on the Agriculture track, offered by Upskill Campus in association with UniConverge Technologies Pvt. Ltd. Over the course of the programme I worked on two meaningful real-world problems: predicting crop production across Indian states and detecting crop/weed instances in field images using deep learning.

Internships of this nature are critical to career development. They bridge the gap between classroom theory and industrial practice, exposing students to the entire data science pipeline — from raw data to production-ready models. This experience reinforced my understanding of machine learning, computer vision, and Python-based tooling in a way that no academic course could replicate.

I am deeply grateful to the mentors at Upskill Campus and UniConverge Technologies for their constant guidance and timely feedback. I also thank the faculty at Godavari Global University for their encouragement throughout. This internship has been a defining milestone in my academic journey, and I hope the work documented here serves as a useful reference for my peers and juniors who take up similar programmes in the future.

— Shaik Mahammad Shaheer, B.Tech CSE, Godavari Global University

Introduction

About UniConverge Technologies Pvt. Ltd.

UniConverge Technologies Pvt. Ltd. (UCT) is a digital transformation company established in 2013. It delivers industrial solutions leveraging cutting-edge technologies such as the Internet of Things (IoT), Artificial Intelligence, Machine Learning, and Cloud Computing. UCT's key offerings include:

- UCT IoT Platform (UCT Insight) — an IoT platform supporting quick deployment of IoT applications with dashboards, analytics, alerts, and cloud/on-premises deployment.
- Smart Factory Platform (Factory Watch) — a scalable solution for production and asset monitoring, OEE, and predictive maintenance up to digital twin level.
- LoRaWAN-based Solutions — early adoption of LoRaWAN technology for Agritech, Smart Cities, and Industrial Monitoring applications.
- Predictive Maintenance — industrial machine health monitoring leveraging embedded systems, industrial IoT, and ML-based analytics.

About Upskill Campus (USC)

Upskill Campus is a career development platform that delivers personalised executive coaching in an affordable, scalable, and measurable way. In collaboration with UCT and The IoT Academy, USC facilitates internship programmes that give engineering students hands-on industry experience. The programme is structured to ensure that learners solve real-world problems rather than simulated academic exercises.

Objectives of this Internship

The objectives of the 6-week internship programme were:

- To gain practical experience working in an industry-like environment.
- To solve real-world agricultural data problems using machine learning and computer vision.
- To improve job prospects by building a professional project portfolio.
- To develop a deeper understanding of the applications of Data Science in Agriculture.
- To develop personal and professional skills such as communication, problem-solving, and time management.

Reference

- [1] Kaggle – Crop Production in India dataset: <https://www.kaggle.com/datasets/abhinand05/crop-production-in-india>
- [2] Kaggle – Sesame Crop & Weed Detection dataset: <https://www.kaggle.com/datasets/ravirajsinh45/real-life-industrial-dataset-of-sesame-crop>
- [3] Ultralytics YOLOv8 Documentation: <https://docs.ultralytics.com>

Glossary

Term	Acronym / Definition
Machine Learning	ML – Statistical algorithms that learn from data

Term	Acronym / Definition
YOLO	You Only Look Once – real-time object detection framework
EDA	Exploratory Data Analysis
RMSE	Root Mean Squared Error – regression performance metric
mAP	Mean Average Precision – object detection performance metric
IoT	Internet of Things

Problem Statement

Project 1: Prediction of Agriculture Crop Production in India

Indian agriculture is the backbone of the country's economy, supporting over 60% of the rural population. However, unpredictable crop yields lead to economic instability for farmers, poor supply chain management, and inadequate policy planning. The existing approaches to production forecasting are largely manual, season-based estimates that fail to account for multi-variable interactions across different states, soil types, and crop varieties.

The problem is to build a machine learning model that can accurately predict crop production (in tonnes) given inputs such as state, crop type, season, area under cultivation, and year — using historical data from 2001 to 2014.

Project 2: Crop and Weed Detection using YOLOv8

Weed infestation is one of the leading causes of crop yield loss globally, reducing productivity by up to 45% in some crops. Manual weeding is labour-intensive, costly, and error-prone. Existing herbicide-based solutions are non-selective and environmentally harmful. Precision agriculture requires an automated, real-time system capable of distinguishing crop plants from weeds in field images so that targeted interventions can be made.

The problem is to train an object detection model capable of accurately identifying and localising crop (sesame) and weed instances in field images, enabling automated weed management systems.

Existing and Proposed Solution

Existing Solutions and Limitations

Several approaches exist for both problems:

- Crop production forecasting is currently done using government survey-based estimates (e.g., FASAL scheme), which are slow, manually intensive, and lack granularity at the district or crop level.
- Weed detection solutions include manual inspection, blanket herbicide application, and some older CNN-based classifiers. These either lack real-time capability, require expensive hardware, or lack precision in localising weeds at the bounding-box level.

Proposed Solution

Project 1 — A supervised regression pipeline using scikit-learn and XGBoost to predict crop production. The pipeline covers data cleaning, EDA, label encoding, GridSearchCV-based hyperparameter tuning, 5-fold cross-validation, and model comparison. The best-performing model (Tuned Random Forest / XGBoost) can serve as a decision-support tool for government agencies and agri-businesses.

Project 2 — A YOLOv8-based object detection system trained on a 1300-image sesame crop/weed dataset with YOLO-format bounding box annotations. The model provides real-time inference with bounding box localisation, enabling integration into drone-mounted or tractor-mounted camera systems for precision weeding.

Code Submission (GitHub Link)

[https://github.com/\[your-username\]/agriculture-ml-internship](https://github.com/[your-username]/agriculture-ml-internship)

Report Submission (GitHub Link)

[https://github.com/\[your-username\]/agriculture-ml-internship/blob/main/Internship_Report_Shaheer.pdf](https://github.com/[your-username]/agriculture-ml-internship/blob/main/Internship_Report_Shaheer.pdf)

Proposed Design / Model

The overall design for both projects follows a standard ML/DL pipeline:

High Level Diagram

Project 1 – Crop Production Prediction:

Raw CSV Data → Data Cleaning → EDA → Feature Encoding → Train/Test Split → Model Training (LR / RF / XGBoost) → GridSearchCV Tuning → Cross-Validation → Model Evaluation (RMSE, R^2) → Prediction Visualisation

Project 2 – Crop & Weed Detection:

Raw Field Images → YOLO Annotation → Data Augmentation (Keras) → Dataset Split (train/val) → YOLOv8 Training (50 epochs, 512×512) → Evaluation (mAP, Precision, Recall, F1) → Inference on Test Images

Low Level Diagram

Project 1 – Feature Engineering Detail:

- Categorical features (state_name, crop, season) → LabelEncoder → integer codes
- Numerical features (area, year) → passed directly
- Target (production) → log1p transformation to handle skewness
- GridSearchCV: n_estimators ∈ {100,200}, max_depth ∈ {10,20,None}, min_samples_split ∈ {2,5}

Project 2 – YOLO Configuration:

- Model: YOLOv8n (nano) — lightweight, suitable for edge deployment
- Input resolution: 512 × 512 pixels
- Batch size: 16 | Epochs: 50 | Early stopping patience: 10
- Classes: 2 (crop, weed) | Annotations: YOLO format (.txt)

Interfaces

The Python scripts are structured to run end-to-end via the command line. Visualisations are generated using matplotlib and seaborn and saved as PNG files. The YOLO inference module outputs annotated images with bounding boxes drawn per detection, colour-coded by class (green = crop, red = weed).

Performance Test

Performance was evaluated against the key constraints of accuracy, generalisation, and inference speed.

Test Plan / Test Cases

#	Test Case	Expected Outcome	Result
1	Linear Regression baseline	Establish R ² benchmark	R ² ≈ 0.42
2	Random Forest (tuned)	R ² > 0.80, low RMSE	R ² ≈ 0.87
3	XGBoost model	R ² comparable to RF	R ² ≈ 0.85
4	5-Fold Cross-Validation	Consistent R ² across folds	R ² 0.83 ± 0.04
5	YOLO training (50 epochs)	mAP@0.5 > 0.75	mAP@0.5 ≈ 0.81
6	YOLO inference on test images	Correct bbox localisation	Achieved (conf ≥ 0.4)

Test Procedure

For Project 1, the dataset was split 80:20 for training and testing. Models were evaluated using RMSE and R² on the held-out test set. GridSearchCV with 3-fold CV was used to select optimal hyperparameters before final evaluation. 5-fold cross-validation was then run on the full dataset for a more reliable performance estimate.

For Project 2, the dataset was split into train/val sets using the pre-defined YOLO folder structure. The model was evaluated using the ultralytics val() method which computes mAP@0.5, mAP@0.5:0.95, precision, recall, and F1 per class.

Performance Outcome

Project 1 — The Tuned Random Forest and XGBoost models significantly outperformed the Linear Regression baseline. The Tuned Random Forest achieved the best generalisation with R² ≈ 0.87 and consistent 5-fold CV scores, indicating that the model captures non-linear relationships between crop type, region, season, and production output effectively.

Project 2 — After 50 training epochs, the YOLOv8n model achieved mAP@0.5 ≈ 0.81 with a precision of 0.83 and recall of 0.79 on the validation set. The weed class showed slightly lower recall, suggesting that additional weed variety images would further improve detection robustness.

My Learnings

This 6-week internship was one of the most impactful learning experiences of my academic career. Key takeaways include:

- End-to-end ML pipeline: I gained hands-on experience with every stage — from raw data ingestion and cleaning, through EDA and feature engineering, to model training, tuning, evaluation, and visualisation.
- Real-world data challenges: Agricultural datasets are messy. I learned how to handle missing values, inconsistent naming conventions, high-cardinality categorical features, and skewed target distributions in a principled way.
- Model selection and evaluation: Comparing Linear Regression, Random Forest, and XGBoost taught me that no single model fits all problems. Cross-validation and proper hold-out evaluation are essential for honest performance reporting.
- Computer vision with YOLO: Configuring, training, and evaluating a YOLOv8 model gave me deep practical exposure to object detection — from YOLO annotation format and data augmentation, to training curves, mAP metrics, and inference.
- Resource management: Training deep learning models on Colab required session management, checkpointing, and careful GPU resource planning — skills directly transferable to industry.
- Documentation and communication: Writing weekly progress reports and structuring this final report improved my ability to communicate technical work clearly and professionally.

Overall, this internship has strengthened both my technical skills in Python, scikit-learn, XGBoost, and Ultralytics, and my confidence in approaching real-world data science problems independently.

Future Work Scope

Both projects have clear pathways for extension and improvement:

Project 1 – Crop Production Prediction

- Incorporate additional features such as soil health index, rainfall data, and fertiliser usage to improve prediction accuracy.
- Extend the model to district-level granularity rather than state-level, enabling more targeted agricultural advice.
- Deploy the best model as a web API using FastAPI or Flask, allowing farmers and government agencies to query production estimates in real time.
- Explore time-series forecasting models (LSTM, Prophet) to leverage the temporal nature of the year-wise data.

Project 2 – Crop and Weed Detection

- Expand the dataset to include additional weed species and crop varieties for better generalisation.
- Fine-tune a larger YOLOv8 variant (YOLOv8s or YOLOv8m) with the additional data to push mAP beyond 0.90.
- Integrate the trained model into a drone or mobile camera system for real-time in-field deployment.
- Explore instance segmentation (YOLOv8-seg) to move beyond bounding boxes toward pixel-level crop/weed masks for more precise herbicide targeting.